MAE 6291
Internet of Things for Engineers

Prof. Kartik Bulusu, MAE Dept.

Week 11 [04/09/2026]	• Localhost • Introduction to Docker: Images and Containers • Guest lecture: IoT and Silicon Security: Dissecting a Real-Life IoT Attack by Chowdary Yanamadala, Technology Strategist, ARM	• Setting up a Docker on the Raspberry Pi 4B • [Graded] In-class lab: Basic containerization process using Docker and the Raspberry Pi 4B

git clone https://github.com/gwu-mae6291-iot/spring2026_codes.git

GW School of Engineering & Applied Science
THE GEORGE WASHINGTON UNIVERSITY

Spring 2026

Photo: Kartik Bulusu

Content Attribution & Sources

This course material incorporates both original content and supplementary visual aids:

- **Images with references:** Properly cited in slide captions, figure legends, or in-text attributions
- **Unreferenced images:** Sourced from Perplexity.io and used for illustrative purposes in an educational context
- **Fair use:** All materials are used in compliance with educational fair use guidelines

Final project proposals are all approved

localhost

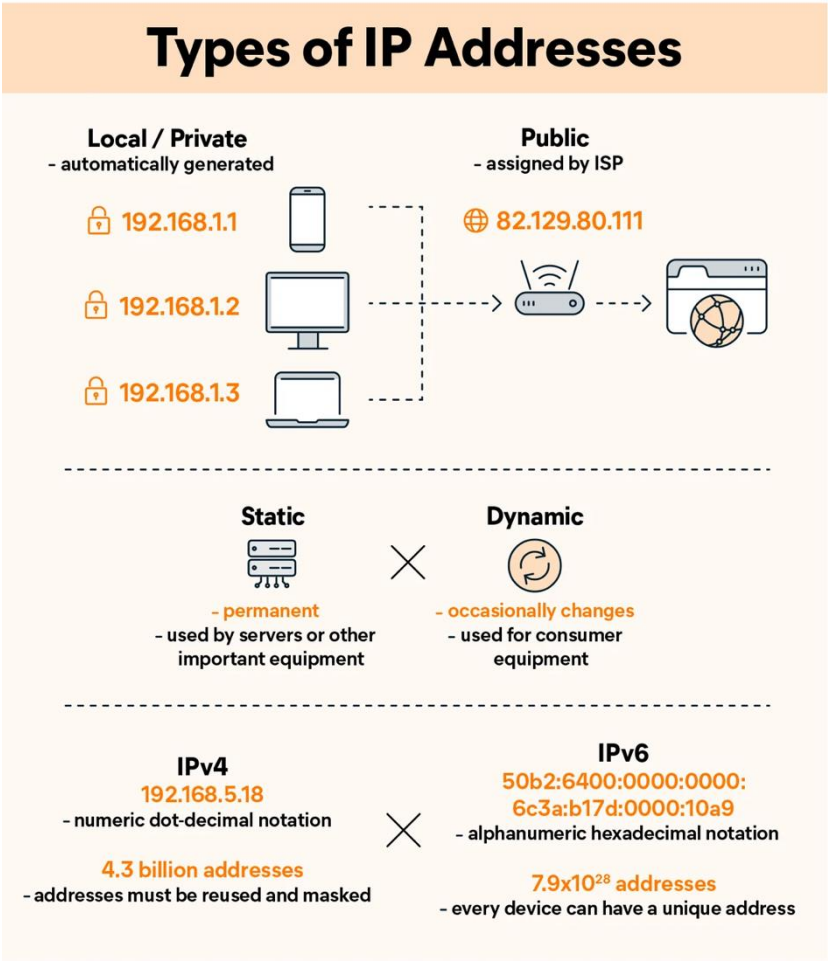
What is an IP address ?

An **IP address** is a unique number that identifies a device on the internet or a local network.

- **A home address** but for computers, helping them find and communicate with each other.
- **A numerical label** assigned to the devices that are connected to the networks, based on the Internet Protocol.

IP addresses come in two versions:

- **IPv4** consists of a string of four 32-bit characters separated by a dot, like **192.168.0.1**; a series of four numbers, ranging from 0 (except the first one) to 255, each separated from the next by a period
- **IPv6** addresses are represented as eight groups of four hexadecimal digits, with the groups separated by colons. A typical IPv6 address might look like this: **2620:0aba2:0d01:2042:0100:8c4d:d370:72b4.**

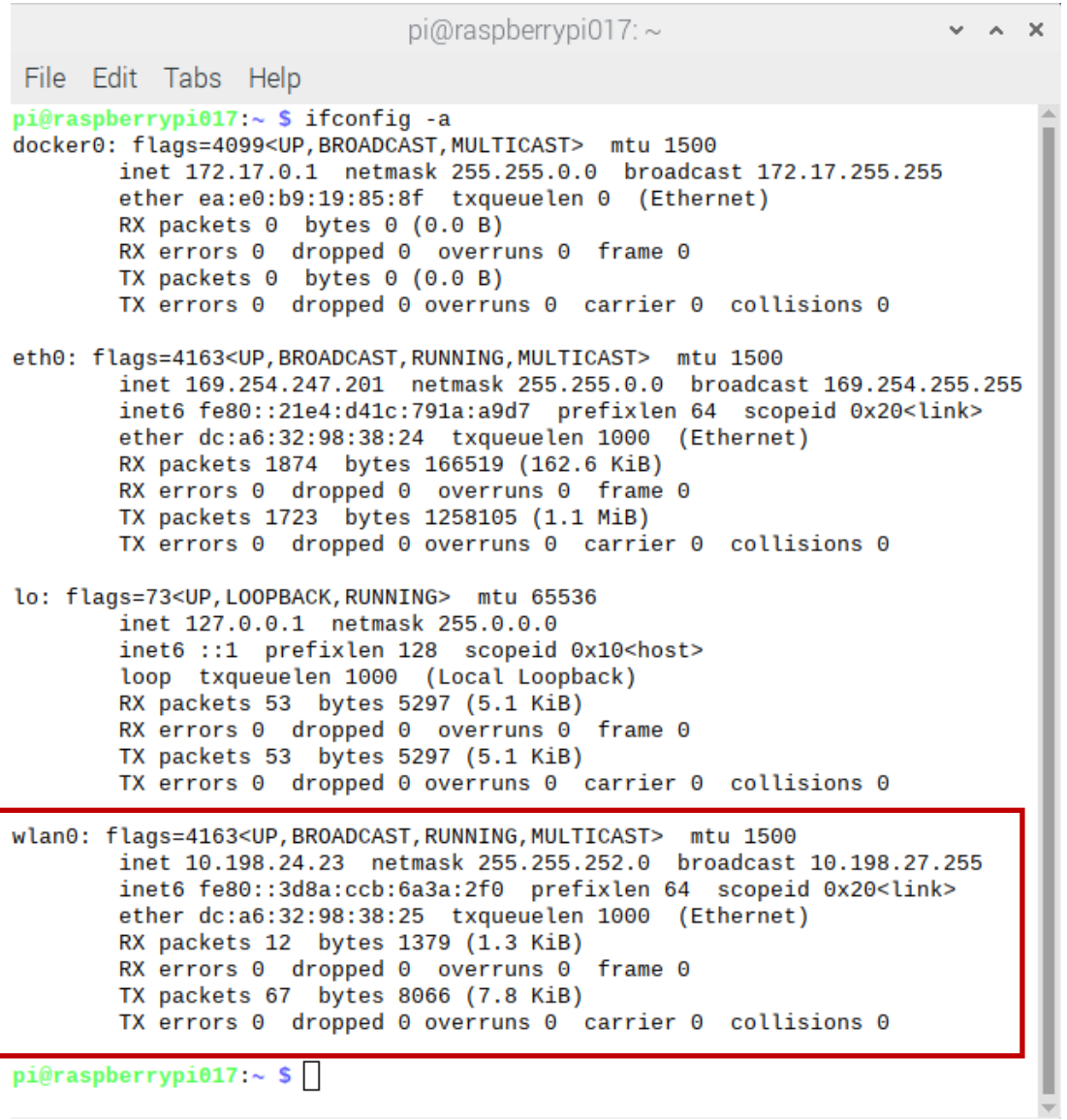


Sources:

Medium: https://medium.com/devgorilla/exploring-localhost-and-127-0-0-1-what-is-the-difference-c109ffce7f29
jocalhost: https://todaybestreports.com/127-0-0-162893-localhost-port-usage-explain/
IP address :https://www.geeksforgeeks.org/what-is-local-host/
https://www.geeksforgeeks.org/open-systems-interconnection-model-osi/
https://www.avast.com/c-what-is-an-ip-address#:~:text=An%20IP%20address%20has%20two,number%20is%20the%20host%20ID.

IP address of your RPi connected to the GWdevice network using WiFi

ifconfig -a

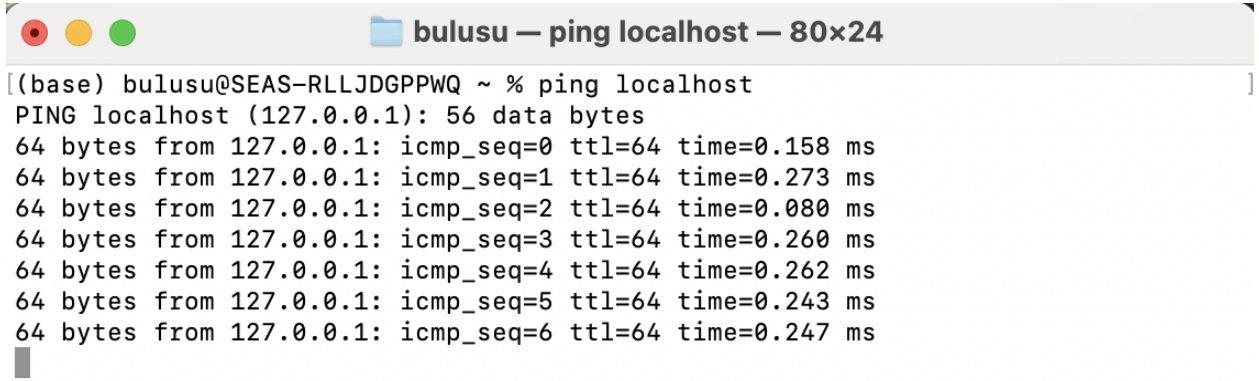


Understanding 127.0.0.1:
The localhost Address

Sources:
Medium: https://medium.com/devgorilla/exploring-localhost-and-127-0-0-1-what-is-the-difference-c109ffce7f29
jocalhost: https://todaybestreports.com/127-0-0-162893-localhost-port-usage-explain/
IP address :https://www.geeksforgeeks.org/what-is-local-host/

The IP address 127.0.0.1 is commonly known as **localhost**.

- It is a special IP address used by a computer to refer to itself.
- Instead of connecting to an external network or the internet, this address is used for loopback communication within the same machine.



On your web browser:
http://localhost
or
http://127.0.0.1

Network Configuration:
A server or program that you are running locally will frequently give you instructions on how to access it via localhost, along with the necessary ports and extra paths.

When you access 127.0.0.1, you're telling your machine to communicate internally rather than looking for an external IP address.

Why do we need a localhost?

Purpose:
The primary purpose of **localhost** is to enable developers

- **To run services locally** for testing without exposing the application to the wider network and
- **To facilitate debugging**, testing web applications, and running local servers.

Universality:
127.0.0.1 is configured as the **localhost** address across all systems.

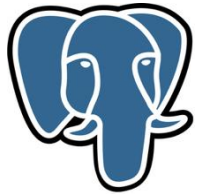
Loopback Address:

The 127.0.0.1 address belongs to the **loopback network**, a reserved IP range used for network diagnostics and testing purposes.

- It allows the machine to send and receive data packets to itself without going through the network interface.

Development Servers

Database Connections




API testing

Browser Debugging

Application Logs

Sources:
Medium: https://medium.com/devgorilla/exploring-localhost-and-127-0-0-1-what-is-the-difference-c109ffce7f29
jocalhost: https://todaybestreports.com/127-0-0-162893-localhost-port-usage-explain/
IP address :https://www.geeksforgeeks.org/what-is-local-host/
NodeJs: By Ryan Dahl, MIT, https://commons.wikimedia.org/w/index.php?curid=26936716
PostgreSQL: By Daniel Lundin - https://wiki.postgresql.org/images/a/a4/PostgreSQL_logo.3colors.svg
By Vectorised from https://labs.mysql.com/common/logos/mysql-logo.svg, Fair use, https://en.wikipedia.org/w/index.php?curid=67634535
By Jamie Dihiansan http://weblog.rubyonrails.org/2016/1/19/new-rails-identity/2 - http://rubyonrails.org/, CC0, https://commons.wikimedia.org/w/index.php?curid=55052527
By The logo is from the following website: www.mongodb.com/brand-resources, Fair use, https://en.wikipedia.org/w/index.php?curid=74919610
By Unknown author - https://www.docker.com/company/newsroom/media-resources/, GPL, https://commons.wikimedia.org/w/index.php?curid=175844098

What is a port in network communication ?

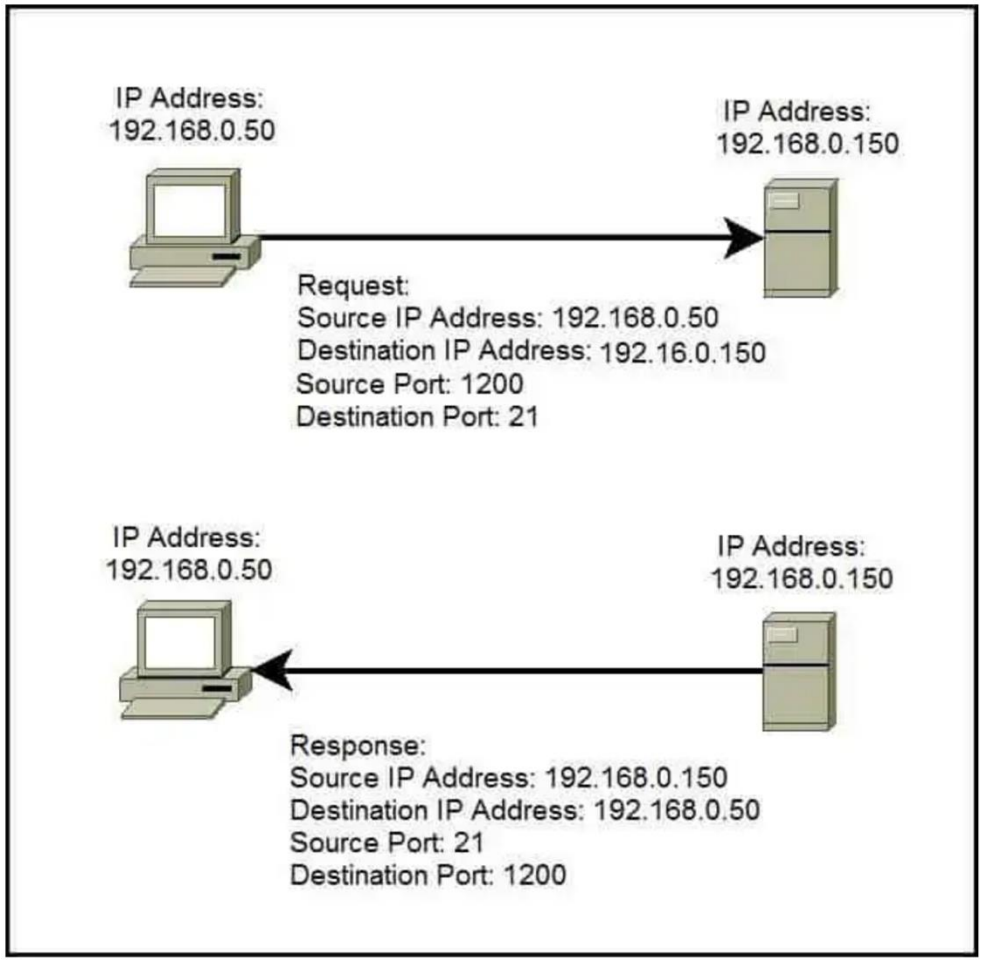
Sources:
Medium: https://medium.com/devgorilla/exploring-localhost-and-127-0-0-1-what-is-the-difference-c109ffce7f29
jocalhost: https://todaybestreports.com/127-0-0-162893-localhost-port-usage-explain/
IP address :https://www.geeksforgeeks.org/what-is-local-host/
https://study-ccna.com/ports-explained/

Definition:
A **port** is a communication endpoint used by software applications to send and receive data over a network.

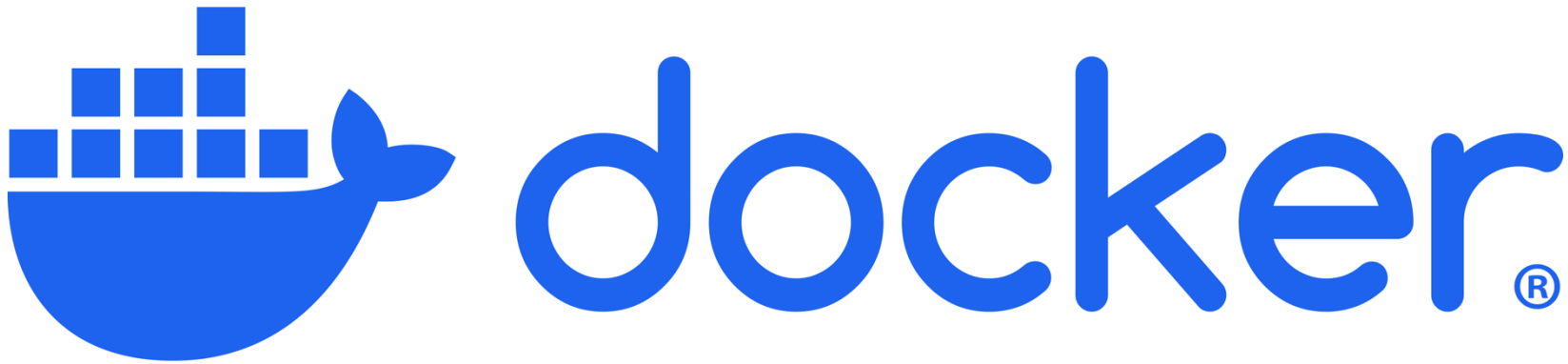
- Ports serve as endpoints for communication, allowing multiple network applications to coexist on a single device.
- Ports are identified by numbers, ranging from 0 to 65535.

Usage:
Different applications use different ports.

- For example, web servers typically use **port 80** for **HTTP** traffic and **port 443** for **HTTPS**.
- In the case of **127.0.0.1:62893**, the number **62893** is likely dynamically assigned by the operating system for a temporary service or local development task.



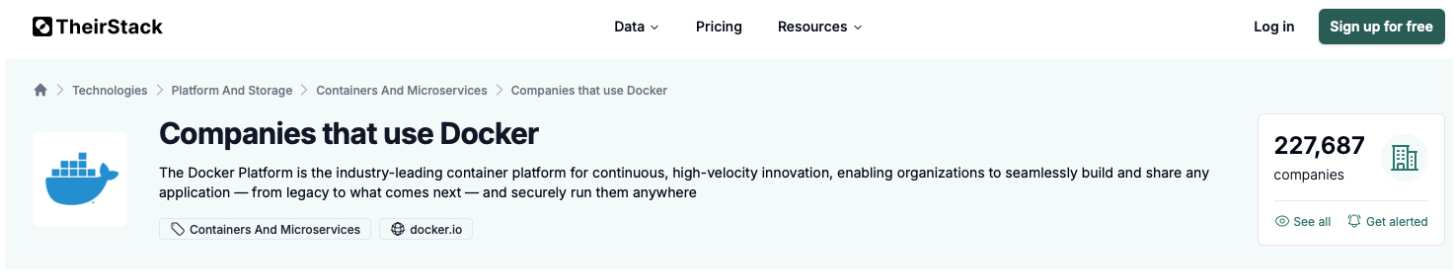
Sources:
https://commons.wikimedia.org/w/index.php?curid=175844098

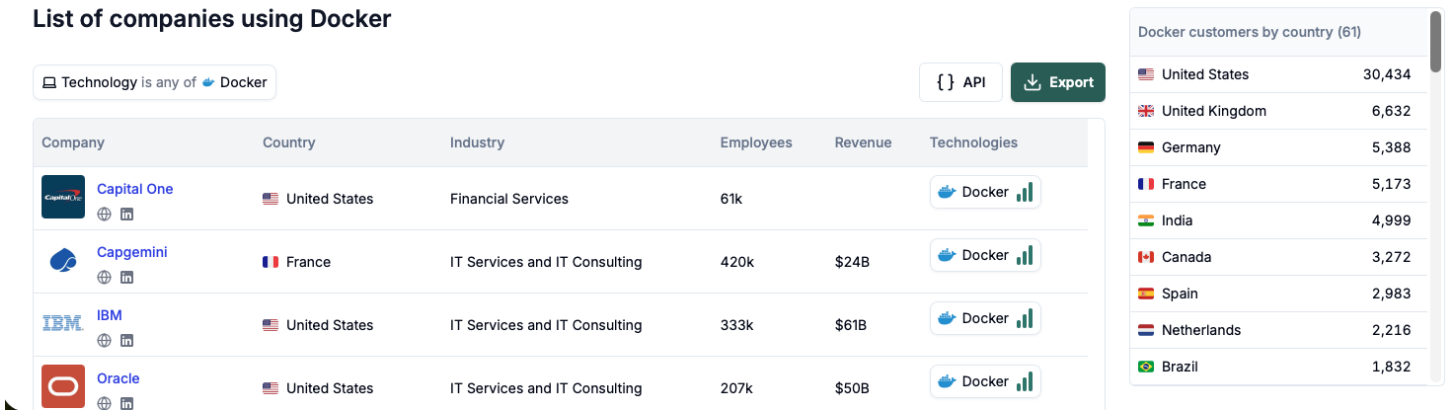


Sources:
https://commons.wikimedia.org/w/index.php?curid=175844098
https://theirstack.com/en/technology/docker

Docker is an open platform for developing, shipping, and running applications, and it uses containers as the unit for distributing and testing software across environments.

- Allows us to package our code
- Allows code to be run on any computer with OS
- Simplifies the process of sharing code



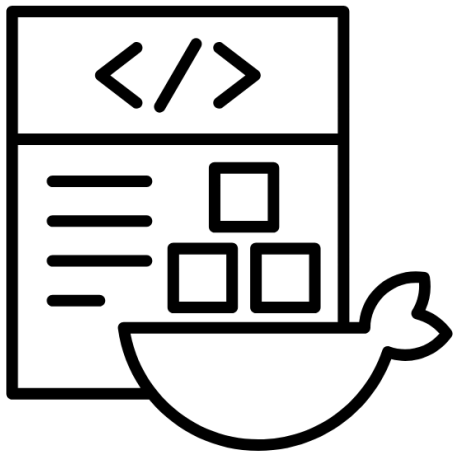


Salary Expectations

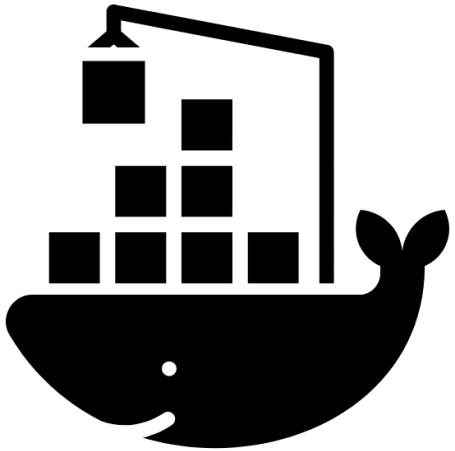
Based on recent 2026 data, salaries for Docker-skilled professionals vary by experience:

- **Entry-Level (0-2 years):** Typically ranges from **$59,900** to **$100,276**.
- **Mid-Level (2-5 years):** Averages around **$108,051**.
- **Senior/Lead (5-10+ years):** Top earners can make between **$130,000** and **$198,000+**.
- **Specialized High-End:** Senior roles at companies like OpenAI or SpaceX can reach total compensation of **$250,000 to $490,000**. Remote Rocketship +2

Sources:
https://commons.wikimedia.org/w/index.php?curid=175844098
Docker Engine by Ali Nur Rohman from https://thenounproject.com/browse/icons/term/docker-engine/, Noun Project (CC BY 3.0)
docker by Afianroc from https://thenounproject.com/browse/icons/term/docker/, Noun Project (CC BY 3.0)



docker®



Docker Images

Contains:

1. Technologies or software we need
2. Runtimes
3. Tools and instructions need to run our code

Docker Containers

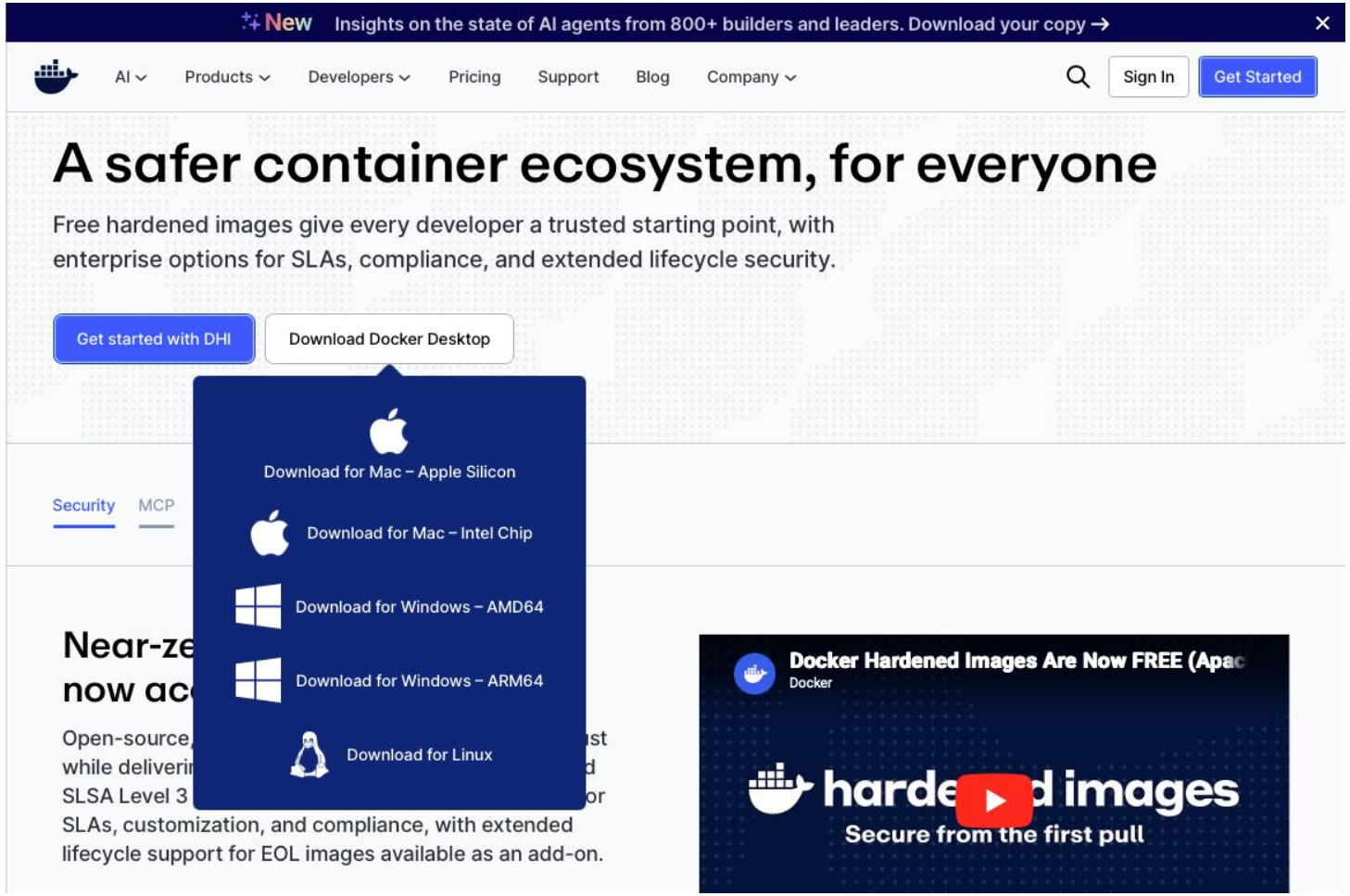
- Runnable instance with an image
- Multiple containers can be created using one image

- Allows us to package our code
- Allows code to be run on any computer with OS
- Simplifies the process of sharing code

Sources:
https://commons.wikimedia.org/w/index.php?curid=175844098
https://theirstack.com/en/technology/docker

Step 0: Install Docker

https://www.docker.com



Sources:
https://commons.wikimedia.org/w/index.php?curid=175844098
https://theirstack.com/en/technology/docker

Step 1: Check if you have Docker

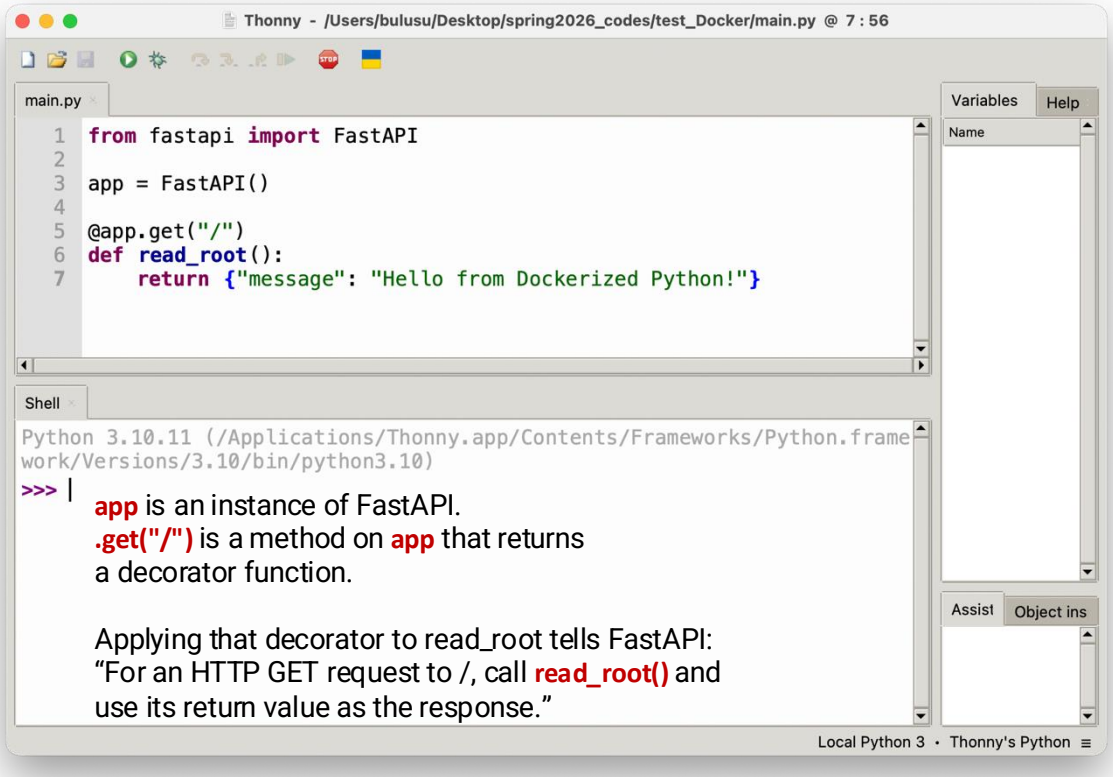
docker --version

docker run hello-world

```
test_Docker — -zsh — 82×29

(base) bulusu@SEAS-RLLJDGPPWQ test_Docker % docker --version
Docker version 29.3.1, build c2be9cc
(base) bulusu@SEAS-RLLJDGPPWQ test_Docker %
(base) bulusu@SEAS-RLLJDGPPWQ test_Docker %
(base) bulusu@SEAS-RLLJDGPPWQ test_Docker %
(base) bulusu@SEAS-RLLJDGPPWQ test_Docker % docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (arm64v8)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

(base) bulusu@SEAS-RLLJDGPPWQ test_Docker %
```

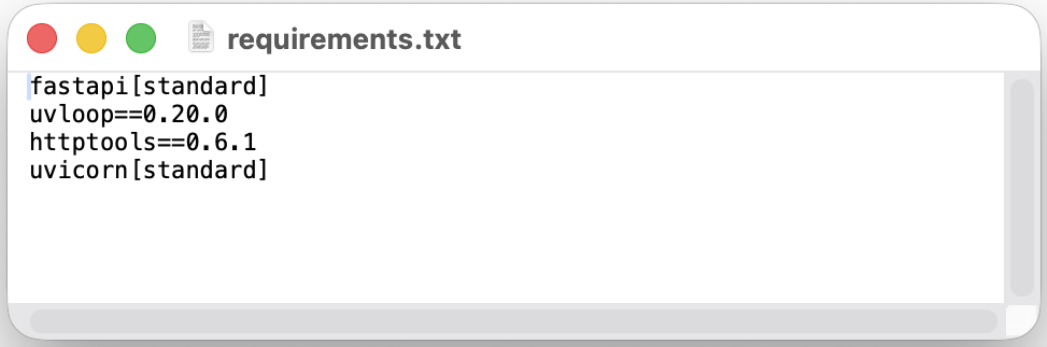
Sources:
https://commons.wikimedia.org/w/index.php?curid=175844098
https://theirstack.com/en/technology/docker

Step 2: Create your Python code and a requirements.txt file



app is an instance of FastAPI.
.get("/") is a method on **app** that returns a decorator function.

Applying that decorator to read_root tells FastAPI: "For an HTTP GET request to /, call **read_root()** and use its return value as the response."



FastAPI is a modern, fast (high-performance), web framework for building APIs with Python based on standard Python type hints.

Uvicorn is a lightning-fast, lightweight ASGI web server for Python, designed for high-performance asynchronous frameworks like ***FastAPI*** and Starlette. It acts as a bridge, accepting HTTP requests and passing them to Python applications using ***uvloop*** and ***httptools*** for speed, supporting *HTTP/1.1* and *WebSockets*.

Save these files in a folder of your choice
Folder for the In-class example: test_Docker

Step 3: Create a DockerFile

Docker Images

Contains:

1. Technologies or software we need
2. Runtimes
3. Tools and instructions need to run our code

Save the “DockerFile” in the same folder as Step 2

```
# syntax=docker/dockerfile:1

## === COMMENT: Tells Docker which Dockerfile syntax/features to use (here, the v1 "modern" syntax).

## === COMMENT: Uses the official python:3.12-slim image as the base, giving you Python 3.12 on a minimal Debian-based image.
FROM python:3.12-slim

## === COMMENT: Sets an environment variable so Python does not create .pyc bytecode files, keeping the image smaller and the filesystem cleaner.
## === COMMENT: Forces Python to run in unbuffered mode so logs are flushed immediately to stdout/stderr (important for seeing logs in Docker).
ENV PYTHONDONTWRITEBYTECODE=1
ENV PYTHONUNBUFFERED=1

## === COMMENT: Set work directory
## === COMMENT: Sets /app as the working directory inside the container; all following relative paths and commands run from here.
WORKDIR /app

## === COMMENT: Install dependencies
COPY requirements.txt .

RUN apt-get update && \
    apt-get install -y --no-install-recommends \
        build-essential \
        gcc \
        libffi-dev \
        libssl-dev \
        python3-dev && \
    rm -rf /var/lib/apt/lists/*

RUN pip install --upgrade pip && \
    pip install --no-cache-dir -r requirements.txt

## === COMMENT: Copy app code
## === COMMENT: Copies all files from your project directory (on the host) into /app in the container, including your application code.
COPY . .

## === COMMENT: Expose port (inside container)
## === COMMENT: Documents that the container listens on port 8000; used by tooling and for clarity, but does not itself publish the port.
EXPOSE 8000

## === COMMENT: Start the app
## === COMMENT: Defines the default command when the container starts: runs Uvicorn to serve the ASGI app app from the main module, binding on all inte
CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
```

Uses the official slim Python 3.12 image as the base

Sets an environment variable so Python does not write .pyc bytecode files to disk and unbuffers Python outputs

Sets /app as the working directory for all following commands

Copies the requirements.txt file from your host into the working directory (/app) inside the image so you can install Python dependencies

This RUN layer installs several system packages and removes cached index files to reduce the final image size

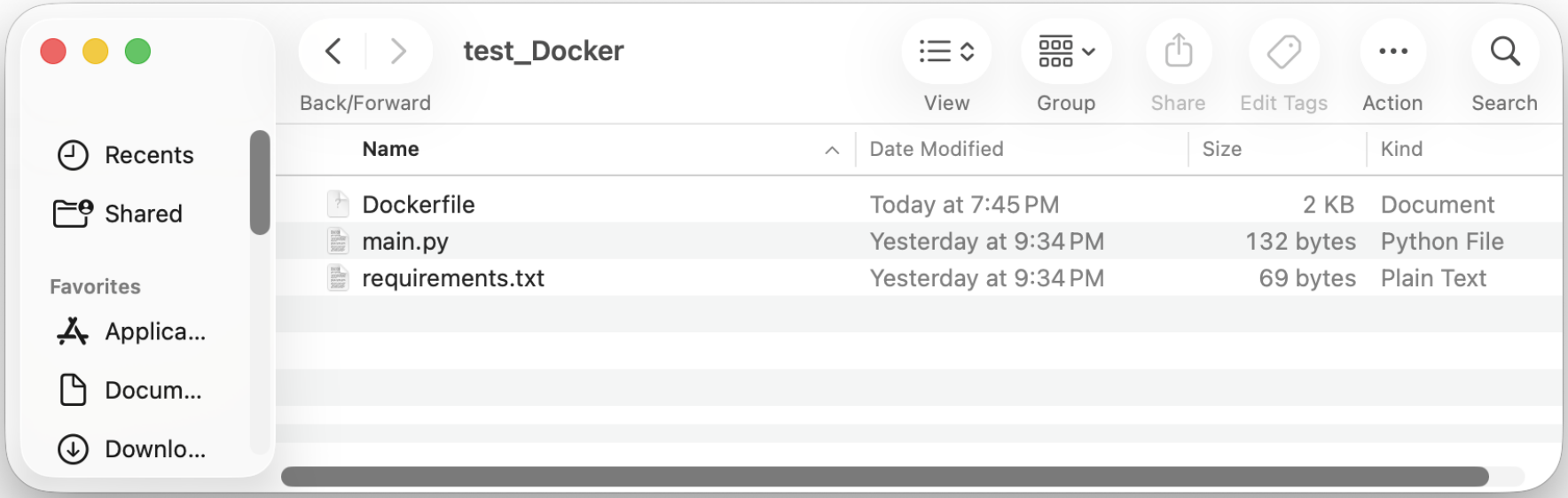
This RUN layer installs pip and upgrades it, and installed Python packages from requirements.txt

Copies everything from the current build context directory on your host into the current working directory in the image (/app)

Documents that the containerized application listens on TCP port 8000

Defines the default command that runs when a container is started from this image

What you have so far …



docker build –t python-app:latest .

python-app is the image name, latest is the tag generated using the option, -t

. tells Docker to use the current directory as the build context

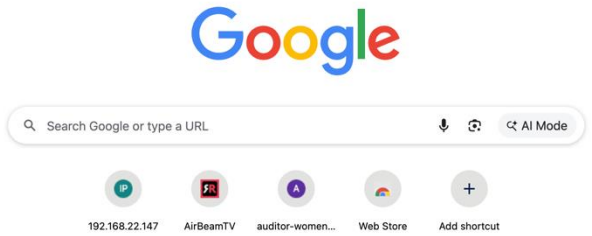
docker run –d –p 8000:8000 --name python-app-container python-app:latest

```
test_Docker — -zsh — 118×6

(base) bulusu@SEAS-RLLJDGPPWQ test_Docker % docker run -d -p 8000:8000 --name python-app-container python-app:latest
49e0c1ea642329267f5dceb281081a14723729cb08b7196b0905b173c612f927
(base) bulusu@SEAS-RLLJDGPPWQ test_Docker %
```

-d runs it in the background.

-p 8000:8000 maps host port 8000 to container port 8000 so you can reach it at http://localhost:8000

--name python-app-container gives the container a friendly name.



Try these and get a graded

docker ps

docker stop python-app-container

docker rm python-app-container

docker images

docker image rm python-app:latest

docker build –t python-app:latest .

docker run –d –p YourIP:8000:8000 --name python-app-container python-app:latest


Photo: Kartik Bulusu
Someone should summarize what we learned today